## 핵심개념

### 1. Scalar Product

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta$$

KEY

- $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$ **(communitative)** and $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$ **(distributive)**
- $\vec{a} \cdot \vec{b} = 0$ if $\vec{a}$ and $\vec{b}$ are **perpendicular**.
- $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|$ if $\vec{a}$ and $\vec{b}$ are **parallel**. In particular, $\vec{a} \cdot \vec{a} = |\vec{a}|^2$

Ex Show that $i \cdot i = 1$ and $i \cdot j = 0$ (*i*, *j*, *k*: unit vectors)

Ex Hence show that $\begin{pmatrix}1\\2\\3\end{pmatrix} \cdot \begin{pmatrix}3\\2\\1\end{pmatrix} = (i+2j+3k) \cdot (3i+2j+k) = 3+4+3 = 10$ (column matrix)

KEY $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta = a_1b_1 + a_2b_2 + a_3b_3$

Ex Given $\vec{a} = (-i+j+3k)$ and $\vec{b} = (7i-2j+2k)$, find the angle between $\vec{a}$ and $\vec{b}$.

Ex Given $\vec{a} = (2i-6j+k)$ and $\vec{b} = (5i+2j+\lambda k)$ are perpendicular, find the value of $\lambda$.

### 2. Line Equation in 3D

| Equation Types | 2D | 3D |
|---|---|---|
| **Vector** | $\vec{r} = \vec{a} + \lambda\vec{b}$<br>$= \begin{pmatrix}a_1\\a_2\end{pmatrix} + \lambda\begin{pmatrix}b_1\\b_2\end{pmatrix}$<br>**<Figure>** | $\vec{r} =$ |
| | **Note:** Position vectors $\vec{a}$<br>Direction vectors $\vec{b}$<br>Parameter $\lambda$ | **Note:** |
| **Parametric** | $\begin{cases}x = a_1 + \lambda b_1\\y = a_2 + \lambda b_2\end{cases}$ | $\begin{cases}x =\\y =\\z =\end{cases}$ |
| **Cartesin** | $\dfrac{x-a_1}{b_1} = \dfrac{y-a_2}{b_2}$<br>$y - a_2 = \dfrac{b_2}{b_1}(x-a_1)$ | |
| | **Note:** $cf.\ y = mx + c$ | **Note:** |

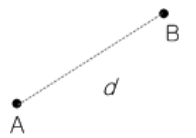
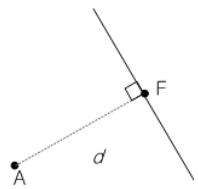
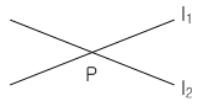
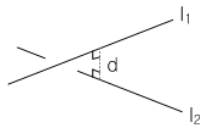
Ex Find a vector equation of the straight line which passes through the point $A(3,4,4)$ and is parallel to the vector $(7i-4k)$

Ex Find a vector equation of the straight line which passes through the point $A(4,5,-1)$ and the point $B(6,3,2)$

Ex Express in a parametric form the coordinates of a point that lies on the straight line
$\vec{r}=(3i+2j-5k)+t(i-6j-2k)$

Ex Find a Cartesian equation of the straight line which passes through the point $A(2,1,9)$ and the point $B(4,1,-8)$

## 3. Types of questions

| | Point | Line | |
|---|---|---|---|
| **Point** |  ① distance | | |
| **Line** |  ② distance |  | ③ intersection point<br>④ angle |
| | |  | ⑤ distance **(FP)** |

Ex Find the distance between two points, whose position vectors are $\vec{a}=(5i-2j)$ and $\vec{b}=(4i-j+3k)$ respectively.

Ex Find the position vector of the intersection point between the two lines:

$$l_1: \vec{r}=\begin{pmatrix}3\\8\\-2\end{pmatrix}+\lambda\begin{pmatrix}2\\-1\\3\end{pmatrix}$$

$$l_2: \vec{r}=\begin{pmatrix}7\\4\\3\end{pmatrix}+\mu\begin{pmatrix}2\\1\\4\end{pmatrix}$$

Ex Find the **acute angle** between the lines with vector equations

$$l_1: \vec{r}=\begin{pmatrix}2\\1\\1\end{pmatrix}+\lambda\begin{pmatrix}3\\-8\\-1\end{pmatrix}$$

$$l_2: \vec{r}=\begin{pmatrix}7\\4\\1\end{pmatrix}+\mu\begin{pmatrix}2\\2\\3\end{pmatrix}$$

Ex Find the distance from the point (4, 1, -1) to the line $l: \vec{r}=(4i-4i+3k)+t(i-2j+2k)$

## 기본문제

**01** In $\triangle OAB$, $P$ is the midpoint of $AB$ and $Q$ is the point on $OP$ such that $OQ = \frac{3}{4}OP$. Given that $\overrightarrow{OA} = a$ and $\overrightarrow{OB} = b$, find, in terms of $a$ and $b$:

a) $\overrightarrow{AB}$ b) $\overrightarrow{OP}$ c) $\overrightarrow{OQ}$ d) $\overrightarrow{AQ}$

The point R on OB is such that $OR = k\,OB$, where $0 < k < 1$.

e) Find the vector $\overrightarrow{AR}$

Given that $AQR$ is a straight line:

f) Find the ratio in which $Q$ divides $AR$ and the value of $k$.

**02** Simplify as far as possible:

a) $\vec{a} \cdot (\vec{b} + \vec{c}) + \vec{b} \cdot (\vec{a} - \vec{c})$, given that $\vec{b}$ is perpendicular to $\vec{c}$

b) $(\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b})$, given that $|\vec{a}| = 2$ and $|\vec{b}| = 3$

c) $(\vec{a} + \vec{b}) \cdot (2\vec{a} - \vec{b})$, given that $\vec{a}$ is perpendicular to $\vec{b}$

**03** Find the point of intersection of the line through the points $(2,0,1)$ and $(-1,3,4)$ and the line through the points $(-1,3,0)$ and $(4,-2,5)$. Calculate the acute angle between the two lines.

**04** With respect to an origin *O*, the position vectors *L*, *M* and *N* are $(4i+4j+7k)$, $(i+3j+2k)$ and $(2i+4j+6k)$ respectively. Find an angle $\angle LMN$.

**05** Find a vector that is perpendicular to both $(2i+j-k)$ and $(i+j-2k)$